

Chen Liu

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Summary

Software Engineer with **over 6 years of experience** building **enterprise-scale applications, distributed systems, and cloud-native platforms.**

Education

New York University

M.S. IN INFORMATION SCIENCE

New York City Jan 2021 - May 2023

Kent State University

B.S. IN COMPUTER SCIENCE

OHIO Jan 2012 - Aug 2015

Experience

InsightFinder AI

SENIOR SOFTWARE ENGINEER

Remote

Nov 2025 - Present

- Designed and implemented **distributed AIOps microservices** for large-scale incident intelligence, enabling automated anomaly detection, alert correlation, and root cause analysis (RCA) across cloud-native systems.
- Developed **observability and analytics pipelines** for RCA confidence scoring, prediction accuracy, system reliability metrics, and operational insights to support production troubleshooting.
- Built deep **ServiceNow integration and platform services**, including incident synchronization, alert workflow automation, feedback ingestion, and data consistency handling across distributed systems.
- Designed and optimized **Cassandra data models** for large-scale time-series and incident data storage, improving query performance and scalability through partitioning and clustering strategies.
- Improved **deployment and operational workflows** using CI/CD pipelines, Docker, Kubernetes, monitoring dashboards, and automation scripts to reduce manual operational effort.
- Designed **vector-based data storage and RAG retrieval pipelines** for operational knowledge systems, focusing on data ingestion, deduplication, scalable querying, and troubleshooting knowledge retrieval.
- Skills:** Spring Boot, Kafka, Redis, Cassandra, MariaDB, Docker, Kubernetes, AWS, Linux, TCP/IP, HTTP, DNS, CI/CD, Prometheus/Grafana, Shell/Python

New York University

CAREER NOTE

New York City, U.S.

Aug 2021 - May 2023

- Completed full-time graduate study for M.S. in Information Science.

IBM Global Business Service

MICROSERVICE ARCHITECT/PRINCIPAL CLOUD INFRASTRUCTURE SOFTWARE ENGINEER

Beijing, China

Jan 2019 - Aug 2021

- Developed and operated a **distributed MySQL database platform** on China Construction Bank (CCB) Cloud, supporting SaaS-based database services for multiple banks with over 50 million users in total.
- Supported **cloud infrastructure operations** across compute, storage, database, and application services, focusing on production stability, capacity planning, and service reliability.
- Designed a database **sharding** strategy and implemented related modules, improving QPS performance by 30% while supporting large-scale financial workloads.
- Performed **Linux, network, and database troubleshooting** for production issues, analyzing system logs, connection failures, latency, resource utilization, slow queries, and service dependencies.
- Optimized **Redis caching** strategies to improve application performance and increased the bank's TPS accesses to over 10,000.
- Built and maintained **Kafka and RabbitMQ-based messaging workflows** to support asynchronous processing, fault tolerance, and reliable data delivery across distributed services.
- Finalized the technical architecture of cloud applications and led design and coding teams using **Agile development, DevOps**, and CI/CD-based release practices.
- Worked on **MongoDB** operations including locking, transactions, indexes, sharding, replication, schema design, and performance tuning to improve overall TPS performance.
- Led a team of 6 junior software engineers to build BMW **Data Lake Platform**, integrating multiple data sources and improving large-scale data processing reliability.
- Deployed and managed cloud resources such as **EC2 instances, RDS databases, and S3 buckets** using **CloudFormation**, supporting repeatable infrastructure provisioning and environment consistency.
- Integrated **various data sources** and ensured seamless data flow across the platform, leveraging **EKS, Docker, Kubernetes**, and monitoring practices for robust and scalable deployment.
- Enhanced data processing and operational performance, resulting in a 30% increase in data throughput and a 20% reduction in latency.
- Skills:** Azure, AWS, Linux, TCP/IP, HTTP, DNS, CI/CD, DevOps, Spring Cloud, Redis, RabbitMQ, Kafka, ElasticSearch, ShardingSphere, MongoDB, MySQL, Docker, Kubernetes, EKS, CloudFormation, Oracle

Bank Of America

Dallas , Tx

SENIOR FULL STACK DEVELOPER

Aug 2015 - Dec 2018

- Built and maintained application that scaled to 2 M daily user, communicating with cross-functional teams regarding product and design.
- Implemented **Spring Gateway** distributed transaction design, based on Monolithic applications for Microservices architecture.
- Enhanced the existing functionality to persist messages to **Cassandra DB** in **Kafka**, met the company's requirements for distributed systems, reduced read repair and improved the performance of read and write by 50 percent.
- partnered with a developer to implement RESTful APIs in Spring Boot, enabling analytic team to increase reporting speed by 23% .
- Coordinating and tracking projects for seamless releases using Project Management System **JIRA** and **SharePoint**.
- **Skills:** JIRA, JQuery, Ajax, JavaScript, JPA, Kafka, Oracle, Spring Boot, Cassandra, Oracle, Linux

IBM China Research Lab

Beijing, China

SENIOR BACKEND DEVELOPER/TEAM LEAD

Aug 2009 - Nov 2011

- Led team of 8 junior software engineers undertaking the software development life cycle for 1 bespoke client projects.
- Implemented a GIS system based on Aegis with IBM Server.
- Built a 5,000 Transaction Processing System (TPS) Financial Management Systems, based on Websphere and MQ cluster.
- Upgrading storage for distributed systems, back-end storage for distributed storage.
- Built a Financial Online System for Taikang Life Insurance Company with Framework: JSF + Spring + Hibernate + Ibatis (SSH) FileNet Workflow.
- Built RESTful APIs that served data that handled over 5000,000 concurrent users.
- Designed RESTful API for the system backend provides mapping URL to frontend. Iterated platform for AI City project, collaborating with a group of 5 engineers to create features across the software.
- Using Selenium I built out a unit testing infrastructure for a client application that reduced the number of bugs reported by the client by 10% month over month.
- Tested software for bugs and operating speed, fixing bugs and documenting processes to increase efficiency by 20%.
- Backend testing based on the LoadRunner.
- Developed cross-platform compatible solutions.
- **Skills:** JQuery, Ajax, JavaScript, Hiberante, Mybatis, WebSphere, IBM MQ, Oracle, Spring Framework, AIX, LoadRunner, FileNet, GIS Server

Additional Experience Post Graduation Internship

Bandwidth Inc. (NASDAQ: BAND)

Raleigh, NC

SENIOR SOFTWARE ENGINEER

Jan 2025 - Aug 2025

- Built an **AI platform** integrating Whisper, BERT, and GPT for voice recognition and intent analysis with **Google Voice support**. Deployed on AWS Lambda to scale for millions of daily calls.
- **Skills:** Cloud infrastructure, Spring Cloud, Redis, DynamoDB, Elasticsearch, Docker, Kubernetes, MariaDB

T-Times Share

Remote

STAFF SOFTWARE ENGINEER

June 2024 - Jan 2025

- Built scalable, **component-driven UI architectures in React and TypeScript**, focusing on reusability, modularity, and maintainable code structure across large applications.
- Collaborated closely with the Product Manager to define and refine the product vision. Designed and implemented the deployment architecture on **AWS**, utilizing AWS RDS for relational data management and **DynamoDB** for high-throughput, low-latency NoSQL storage, ensuring scalability and reliability.
- **Skills:** Cloud infrastructure, Spring Cloud, Redis, DynamoDB, Elasticsearch, Docker, Kubernetes, Mysql

PAX Technology

Jacksonville, FL

SENIOR SOFTWARE DEVELOPER

July 2023 - May 2024

- Designed and implemented a **highly available and scalable Payment service** and **E-commerce** platform, enabling **Bank of America's** customers to seamlessly purchase the company's POS products based on **Kotlin, Spring Boot** and **AWS**.
- **Skills:** AWS Infrastructure, Spring Cloud, Redis, DynamoDB, Docker, Kubernetes, PostgreSQL